Sri Atragada

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EDUCATION

Stony Brook University **Stony Brook, New York**
Bachelor of Science in Computer Science, Minor in Finance *Expected May 2028*

RELEVANT EXPERIENCE

Software Engineer Intern **Stony Brook, NY**
Stony Brook University *September 2025 - February 2026*

- Designed and shipped a Python NLP service that converts natural-language library queries into Boolean search expressions, deployed into the SBU Libraries production stack used by 25K+ students/faculty monthly.
- Developed an end-to-end NLP pipeline using Python and OpenAI GPT-4o to extract entities, temporal constraints (e.g., "last 5 years"), and resource facets from raw user input, reducing "zero-result" searches by 35%, optimizing query precision.
- Integrated semantic retrieval over the library catalog using Pinecone + sentence-transformer embeddings, layering it on top of the existing lexical search to surface contextually relevant results that keyword matching missed.

Software Engineer Intern **Portland, ME**
Atlas Legacy Inc. *May 2025 - August 2025*

- Built containerized AWS ECS deployment with GitHub Actions CI/CD pipelines, cutting release cycle time by 40% and eliminating manual deployment errors across staging and production
- Implemented an end-to-end demand forecasting pipeline in Python, ingesting historical sales and inventory data to predict stock needs 30 days out, eliminating 10 hours of manual ops planning per week
- Reduced AWS EC2 costs by 20% by auditing over-provisioned instances using Cost Explorer and Compute Optimizer, recommending right-sized instance migrations across staging and production environments

PROJECTS

Chronicle | React, TypeScript, Python, Vite, Zustand, Gemini API, ChromaDB GitHub

- Architected a fully playable 10,000 × 10,000 tile open-world RPG engine where the player navigates a living world of hundreds of autonomous agents — rival kingdoms, merchant companies, and military factions — each running independent goal-state machines and territorial strategies across a staged multi-phase simulation tick loop
- Developed a Python RAG pipeline to give agents a bounded context window by retrieving semantically relevant events and NPC memories from a ChromaDB vector store, producing historically consistent agent responses at runtime

FormFlow - Code-A-Site Hackathon 2026 2nd Place | JavaScript, Google MediaPipe Pose, MongoDB, Socket.IO GitHub

- Built a real-time AI fitness platform that scores workout form rep-by-rep via webcam and drives live competitive leaderboards across multiplayer sessions using Socket.IO
- Engineered a client-side computer vision pipeline with MediaPipe Pose, offloading inference to the browser; applied EMA smoothing on angular velocity streams to eliminate false-positive form detections under high-intensity movement
- Decoupled state persistence from social event delivery using MongoDB Change Streams, removing the need for a dedicated message broker and enabling horizontal scaling of real-time multiplayer rooms

High-Frequency Order Matching Engine | C++, CMake, GoogleTest GitHub

- Engineered a low-latency matching engine in C++ implementing a Price-Time Priority (FIFO) algorithm and a Pro-Rata allocation model utilizing a largest-remainder split to execute limit and market orders with deterministic sub-microsecond performance.
- Built a CLI research tool with support for JSON-line reporting and multi-level sweep patterns, enabling rapid backtesting and trade analysis via scripted inputs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, R
Frameworks and Libraries: React, Spring Boot, Pinecone, FastAPI, LangChain, ChromaDB
Tools and Platforms: Node.js, MongoDB, Figma, Docker, Amazon Web Services,
Data & ML: PyTorch, scikit-learn, Pandas, NumPy
Databases: PostgreSQL, MongoDB, Redis, Supabase